SDC HACKATHON 2026

Phase 1 Preliminary Project Proposal

Repath | gravity | 25 October 2026

1. Team and Project Overview

Team Name	gravity
Selected Domain	Education and Workforce Development
Selected SDG Track (1–17)	SDG 8 - Decent Work and Economic Growth
Project / Solution Name	Repath
Name (Leader)	Wu Tong
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University Affiliation(s)	Xiamen University Malaysia
Number of Team Members	2 - Wu Tong (Leader); Liang Yiyan Member email: cst2609035@xmu.edu.my

2. Executive Summary and Problem Statement

2.1 The Problem Context

Workers facing career disruption and graduates entering the labour market often struggle to translate their existing experience into suitable roles. Job titles alone do not show transferable skills, while scattered vacancy information and generic course recommendations make it difficult to choose a realistic next step. Limited time and financial pressure can make unnecessary retraining especially costly.

In Malaysia, DOSM reported that 1.6004 million employed graduates experienced skill-related underemployment in 2024, representing 32.2% of employed graduates [1]. This national benchmark establishes the scale of skills mismatch; it is not a measurement of Repath users or an outcome caused by the product.

Repath focuses on Malaysian career changers, displaced workers and graduates seeking a clearer transition plan. It helps users identify capabilities demonstrated in past projects, compare adjacent careers, and select learning activities that fit their circumstances. The social and economic objective is to make career planning more accessible while preserving the value of experience people already have.

2.2 The Proposed AI Solution

Repath is a multilingual career transition application that connects project experience, current circumstances, career options and a practical preparation plan. Users confirm their capabilities, inspect transferable skills and gaps across eight career directions, and choose weekly actions that match their available learning time. English, Chinese and Bahasa Melayu interfaces support broader access.

The current deployed demo uses transparent, browser-based matching rules, local record storage, progress tracking, and JSON backup and restore. It separates personal fit from labour-market evidence and identifies the limits of quarterly Malaysian job-advertisement data and US O*NET occupational references. No live generative-model API is currently connected.

For Phase 2, we propose adding a server-side GenAI service to turn user-approved experience descriptions into suggested skill evidence and draft learning plans. Users will confirm extracted skills before they affect recommendations. Structured output validation, source references and the existing rules-based fallback will reduce unsupported recommendations. Model selection and API access remain to be finalized.

Our team completed the mandatory GenAI, database and cloud deployment workshops and used the organizer-provided Replit environment. We plan to apply those skills to API integration, consent-based data storage and deployment. We have not used a sponsor-provided GenAI API. The final prototype will be evaluated on planning efficiency and recommendation clarity rather than claimed employment outcomes.

3. Technical Architecture and Implementation Plan

3.1 Conceptual Tech Stack

Implemented: HTML, CSS and JavaScript ES modules; a Node.js development server and build scripts; browser localStorage; JSON import/export; and a deployed Sites frontend. The organizer-provided Replit environment was used for development. The current public demo is hosted on Sites, not Replit.
Data: selected O*NET 31.0 occupations and tasks (CC BY 4.0); attributed DOSM quarterly job-advertisement excerpts; linked WEF context; and optional arXiv RSS metadata. These are reference snapshots, not live hiring guarantees.
Phase 2 plan: a Node.js backend will call a selected GenAI API, validate structured responses, and return suggested skills and plan drafts for user confirmation. API credentials will remain server-side. A relational database (proposed SQLite for the pilot) will store consented profiles, plans and progress; schema design and access controls will precede any migration from local-only storage.
Architecture: user experience input -> frontend -> backend validation -> GenAI service -> user confirmation -> explainable matching + reference data -> career options and weekly plan -> local records / planned database.
Sponsor resources: organizer-provided Replit was used. No sponsor GenAI API has been used. GenAI and database components above are planned, not implemented.
Demo: https://repath-ai-0-2.lianghuihu.chatgpt.site

3.2 Design Thinking and Deployment Strategy

Our current design frames career transition around demonstrated experience and practical constraints. The prototype lets users inspect skill gaps and source evidence rather than treating a single score as a hiring prediction. Multilingual interfaces, theme options and responsive layouts support different usage contexts.
We have iterated the working flow from experience entry through career comparison to a weekly plan. Existing automated checks cover matching, planning, source handling, record backup and invalid imports. Browser checks cover mobile layouts and persistence. We have not yet completed formal user interviews or an employment-impact study.
Before Phase 2, we plan to recruit 10 consenting target users, observe their current planning process, test the same planning task with Repath, and revise confusing steps. We will pilot AI suggestions with explicit user confirmation and verify that the fallback works when the model is unavailable. Deployment checks will include secret management, consent, record deletion, accessibility and data-source freshness. The pilot will measure planning time and task completion; job placement and income improvement require longer follow-up.

4. Preliminary Environmental and SDG Impact Framework

Target SDG Sub Targets

Primary: SDG Target 8.5, productive employment and decent work for all, including young people and persons with disabilities [2]. Repath supports career preparation; it does not measure the official national SDG indicator directly.

Current Problem Baseline

Published baseline: 1.6004 million employed Malaysian graduates in skill-related underemployment in 2024 (32.2%) [1]. Pilot planning-time baseline: not yet measured; collect each participant's time using their usual method before the Repath task. Do not substitute the national rate for a product baseline.

Core Measurable KPI

Primary KPI: percentage reduction in median time to produce a usable career plan. Reduction = (baseline median - Repath median) / baseline median x 100%. A usable plan names one career direction, evidence of existing skills, gaps and three specific next actions. Pilot: 10 consenting target users; counterbalance comparable tasks and report completion rates.

Phase 2 Impact Target

Proposed Phase 2 target: at least 20% reduction in median planning time versus the measured baseline. This is a pilot goal, not an achieved result or a forecast of national unemployment reduction. Report sample size and limitations. No carbon or energy savings are claimed.

[1] DOSM, Graduates Statistics 2024:
https://www.dosm.gov.my/uploads/release-content/file_20251031111142.pdf
[2] UN SDG 8 targets: https://sdgs.un.org/goals/goal8

5. Eligibility and Readiness Checklist

[X] All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.

[X] The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.

[X] The proposed solution targets a functional, deployed application rather than a theoretical business case study.

[X] All fields in this template have been completed within the word limits stated.

6. IP and Originality Declaration

6.1 Pre Existing or Prior Owned Work

Does this submission include any work created or owned before the hackathon? [ ] Yes [X] No. None, as confirmed by the team leader.

6.2 Third Party Code Datasets or Models

[X] Yes [ ] No. Third-party materials and services are declared below.

O*NET 31.0 | Occupational dataset | CC BY 4.0; selected records and adapted labels credited in NOTICE.

DOSM Job Market Insights | Statistical excerpts | DOSM source attribution; original rights retained; no full dataset relicensed.

WEF Future of Jobs 2025; arXiv RSS | Linked context / metadata | Original source terms apply; no report or paper full text redistributed.

DM Sans; Noto Sans SC | Fonts | SIL Open Font License; externally loaded, no font binaries bundled.

Node.js | Development runtime | MIT license; third-party notices apply.

OpenAI Codex; Replit; Sites | Development and hosting services | Provider terms apply; no model weights redistributed and no runtime GenAI API currently used.

Microsoft Learn; OpenLearn | Learning-resource links | Original provider terms; course content is linked, not copied.

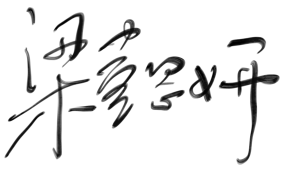
6.3 Employment Related IP Conflicts

Could an employer claim ownership of the hackathon work? [ ] Yes [X] No. None, as confirmed by the team leader.

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team’s knowledge and that the project will be original work created during the SDC 2026 hackathon period.

Team signatures





Date: 25 October 2026